

S7CAS-PLEIADES-CARTE-MERE

Pleiades

01

Revision 1.00

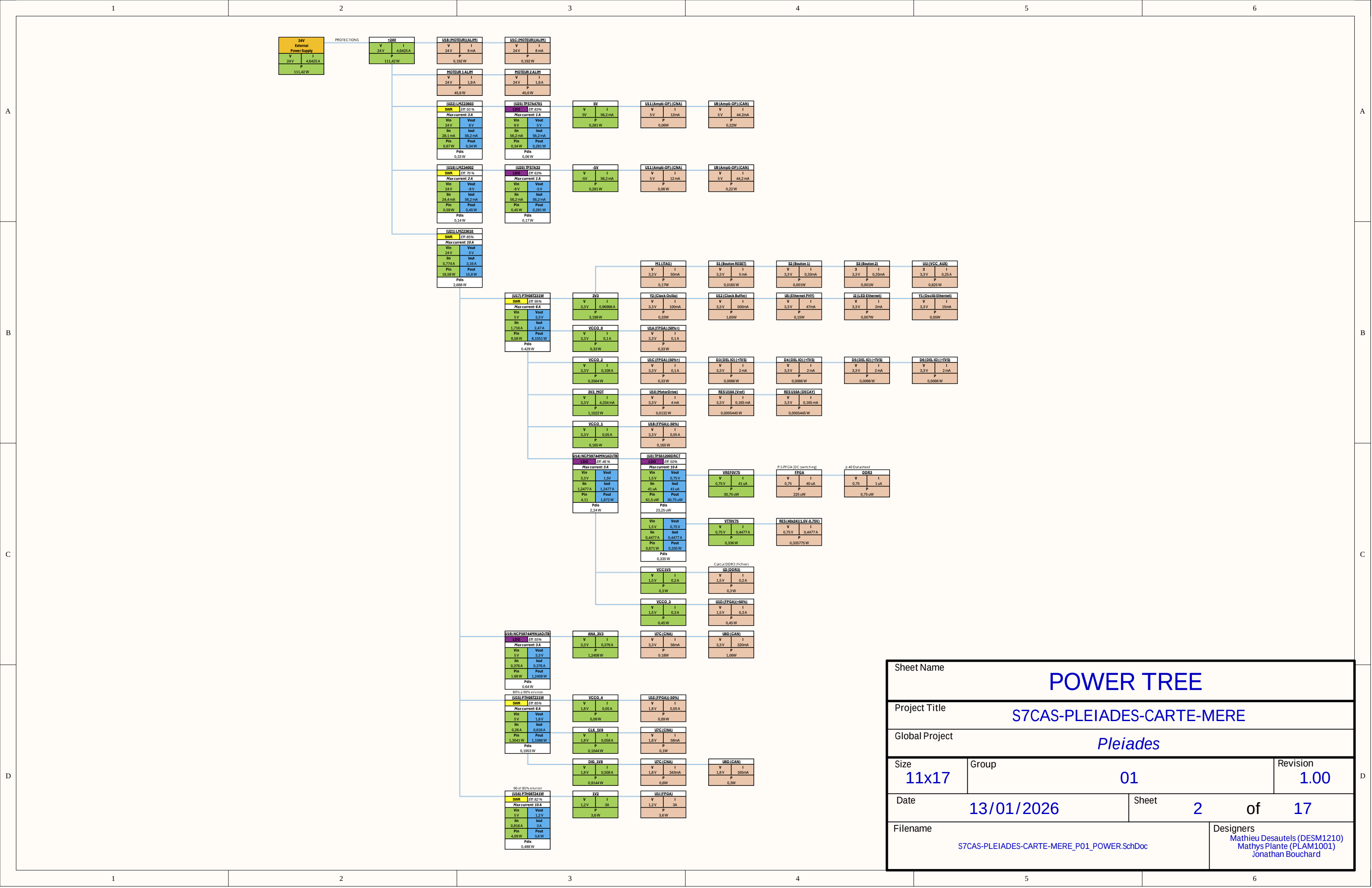
Date : 13/01/2026

Revision history	

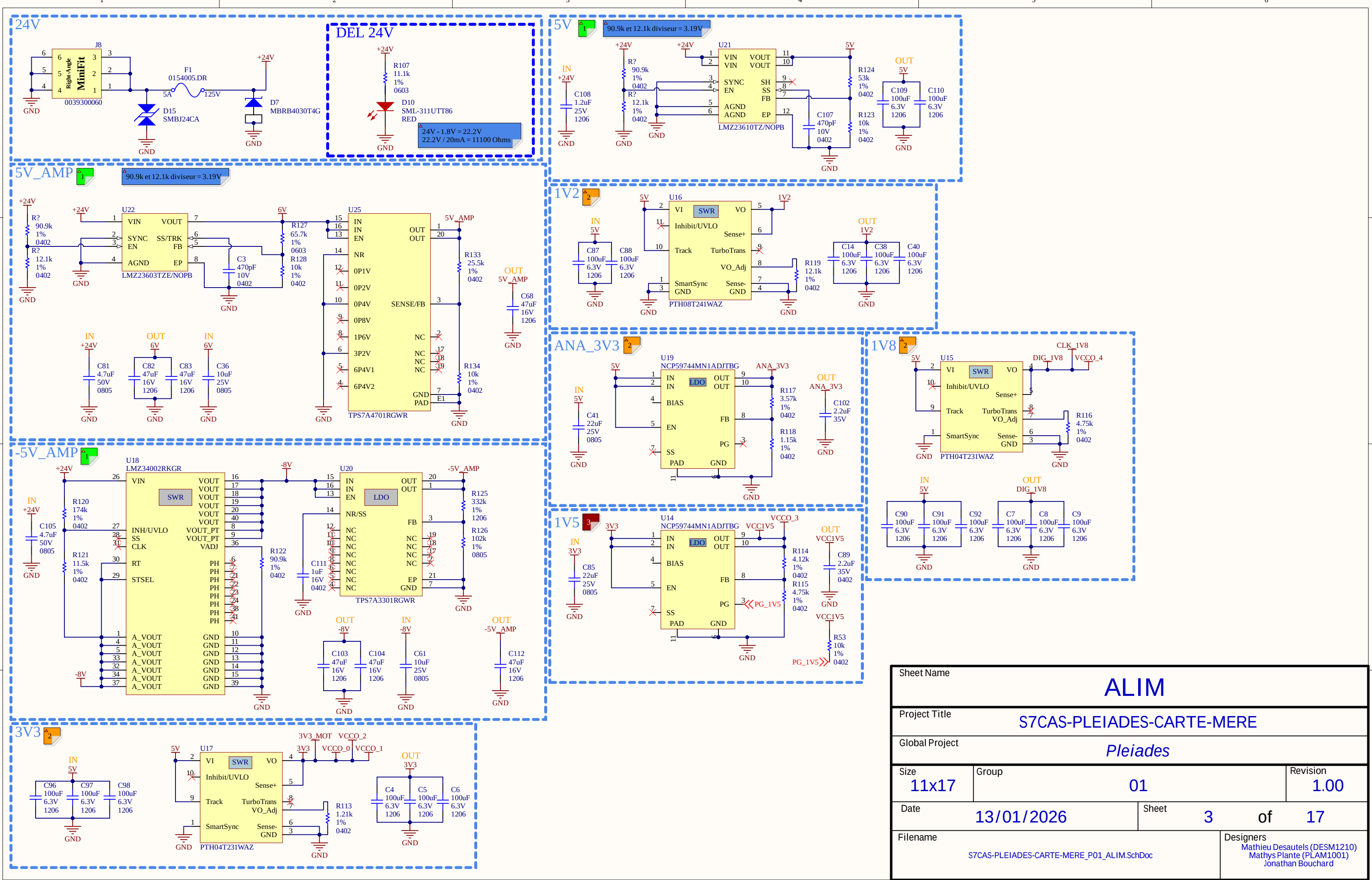
Package size conversion	
Metric	Imperial
1005	0402
1608	0603
2012	0805
3216	1206
3225	1210
6432	2512



Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
Date	13/01/2026	Sheet	1 of 17
Filename		Designers	
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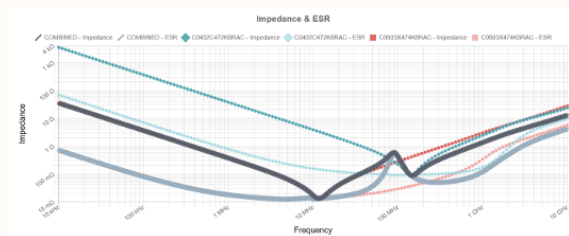
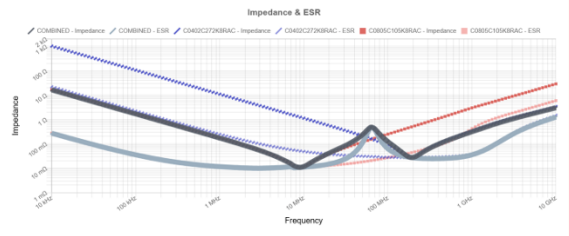
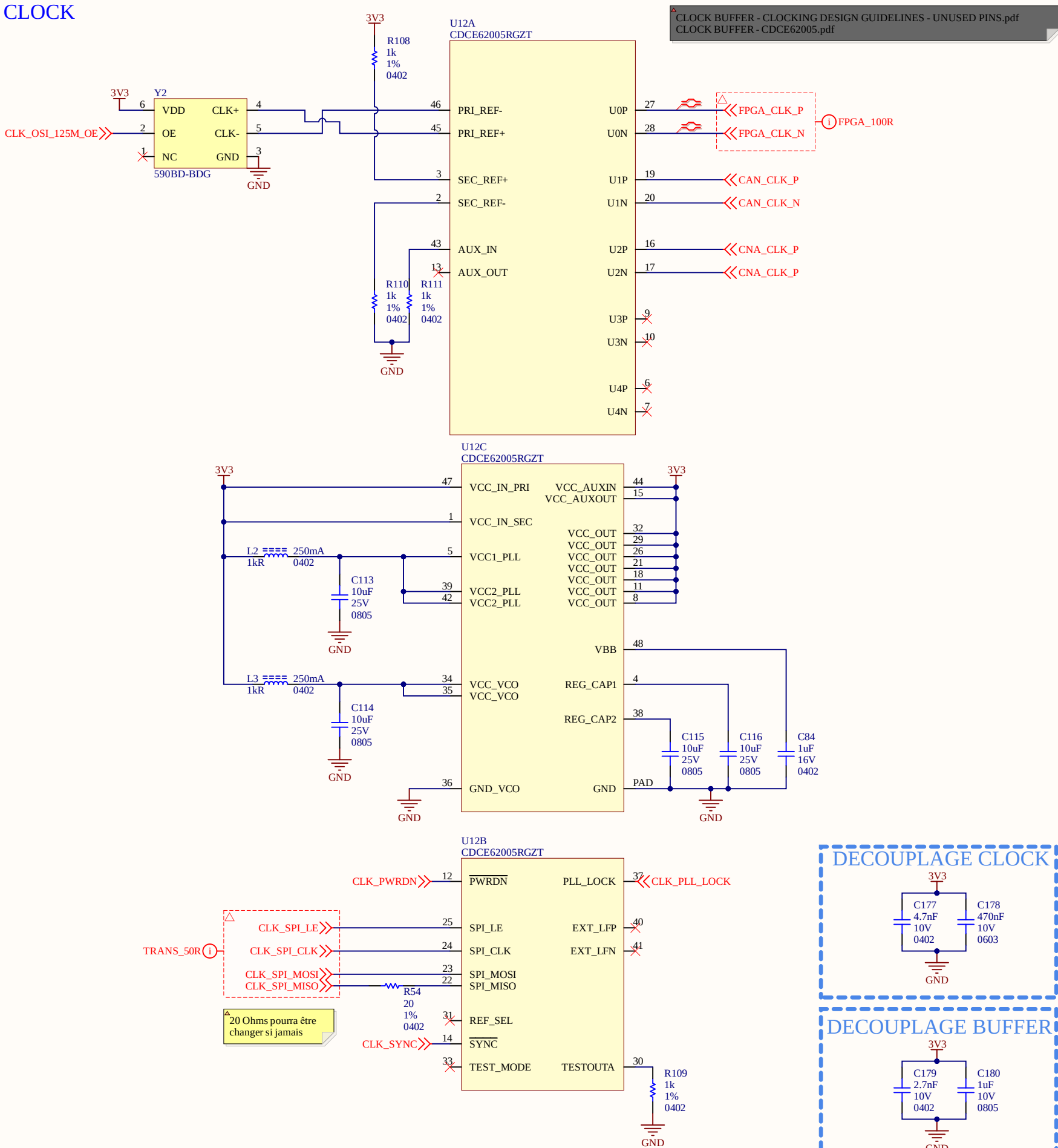


Sheet Name			POWER TREE		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group			Revision	
11x17	01			1.00	
Date	13/01/2026		Sheet	2	of 17
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S7CAS-PLEIADES-CARTE-MERE_P01_POWER.SchDoc				Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard	



Sheet Name		ALIM	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
Size	Group	Revision	
11x17	01	1.00	
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Filename		Designers	
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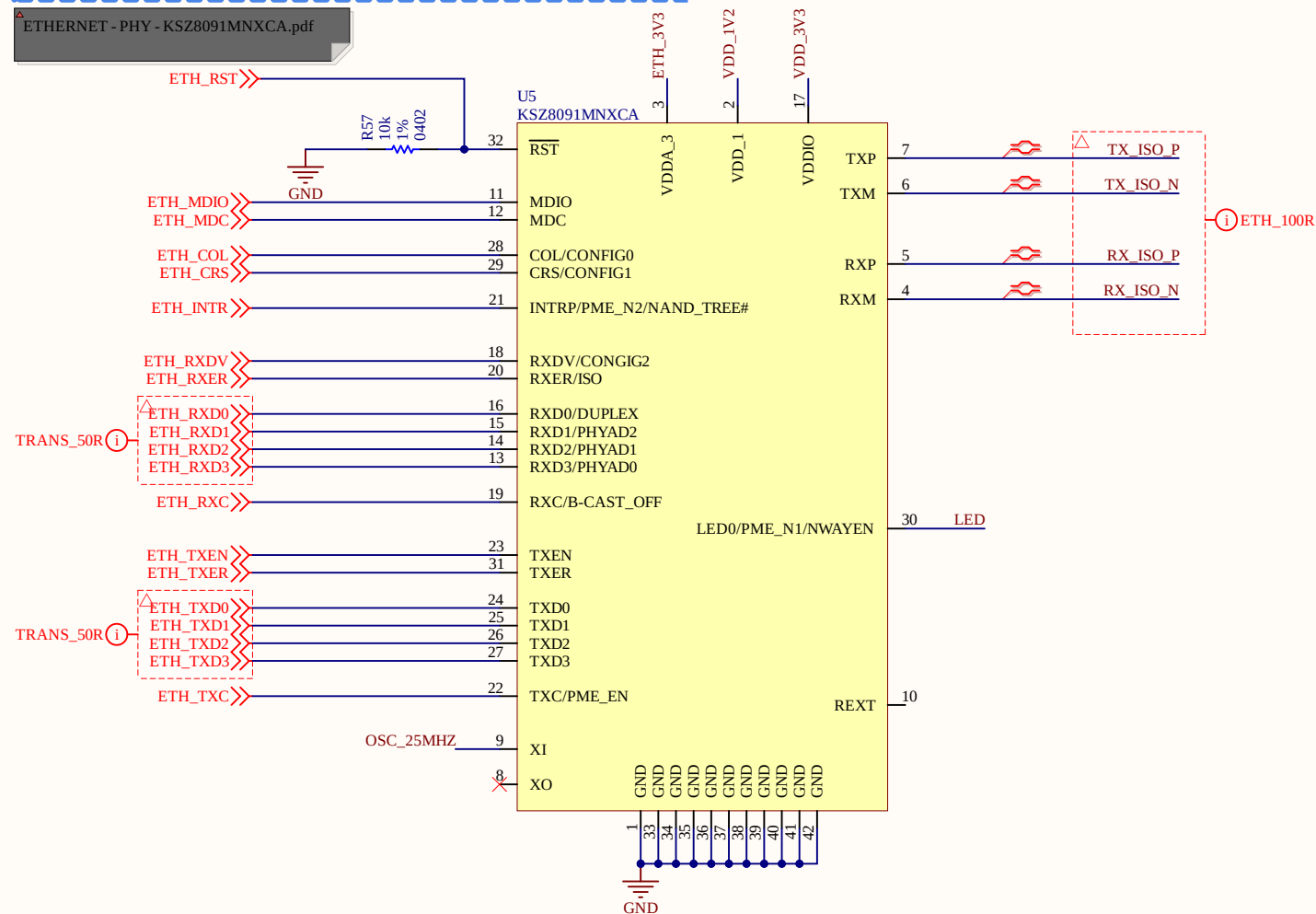
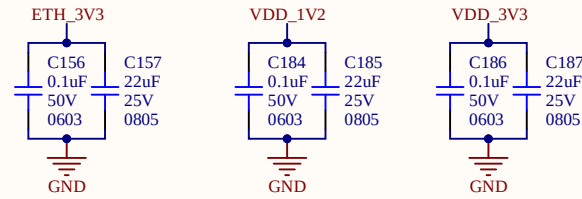
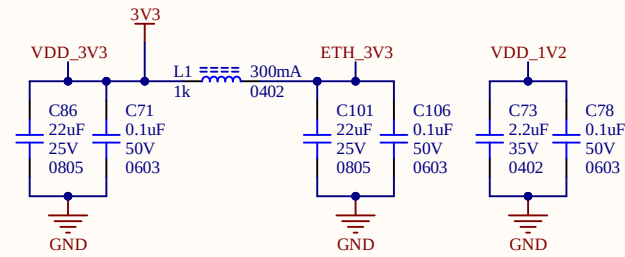
CLOCK



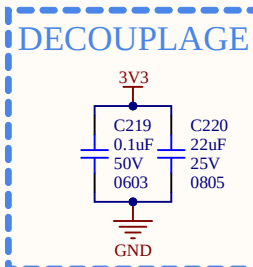
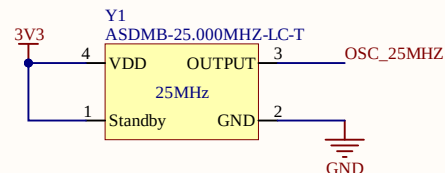
-Quand tu es en LVDS faut que branche le SEC à une résistance (CLOCK BUFFER - CLOCKING DESIGN GUIDELINES - UNUSED PINS.pdf)
-CAN et CNA pas fait pour la pair différentiel puisqu'elle est directement dans le CAN et CNA

Sheet Name			CLOCK		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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11x17	01			1.00	
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_CLOCK.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

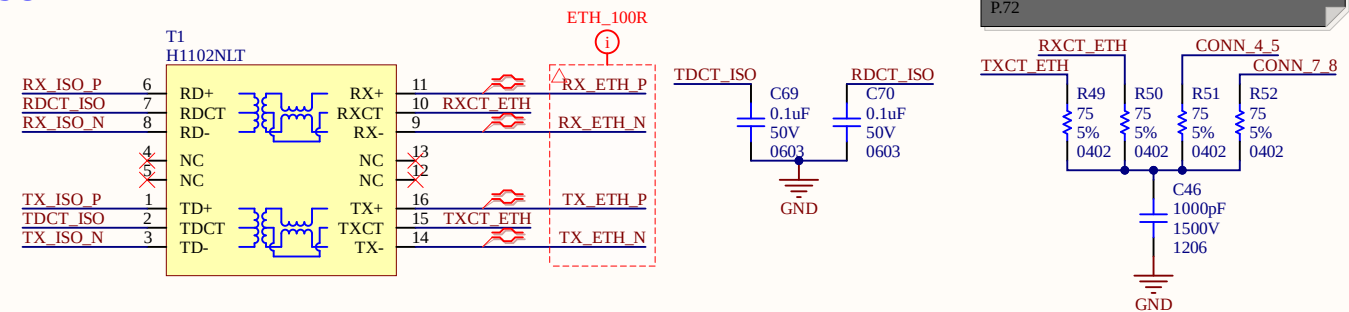
ETHERNET ALIMENTATION



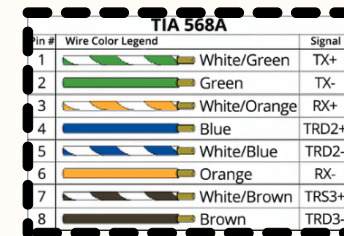
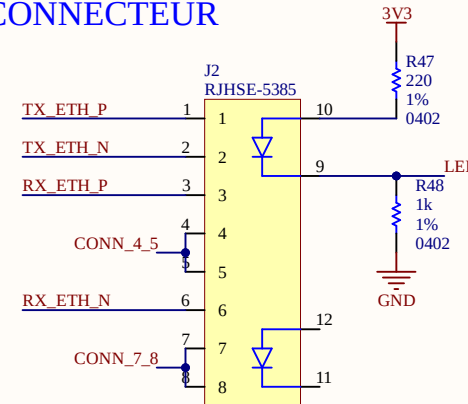
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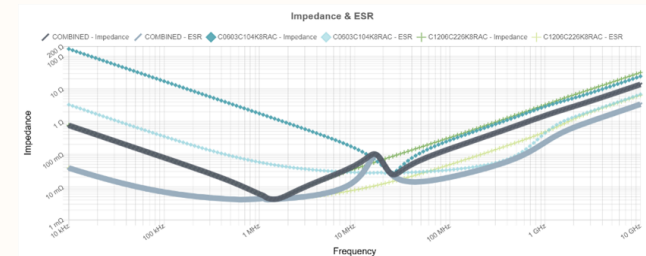
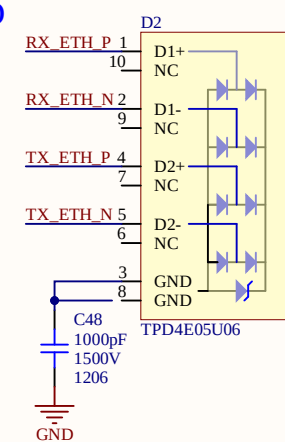
ISO



CONNECTEUR



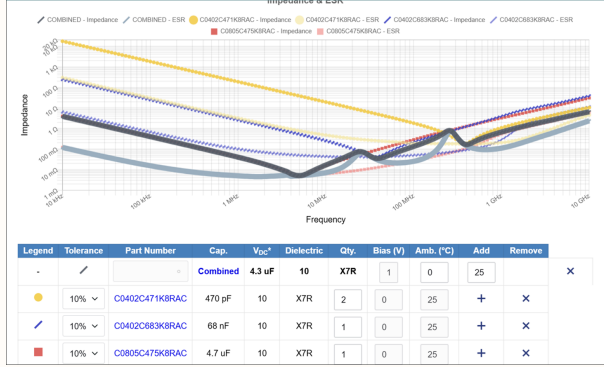
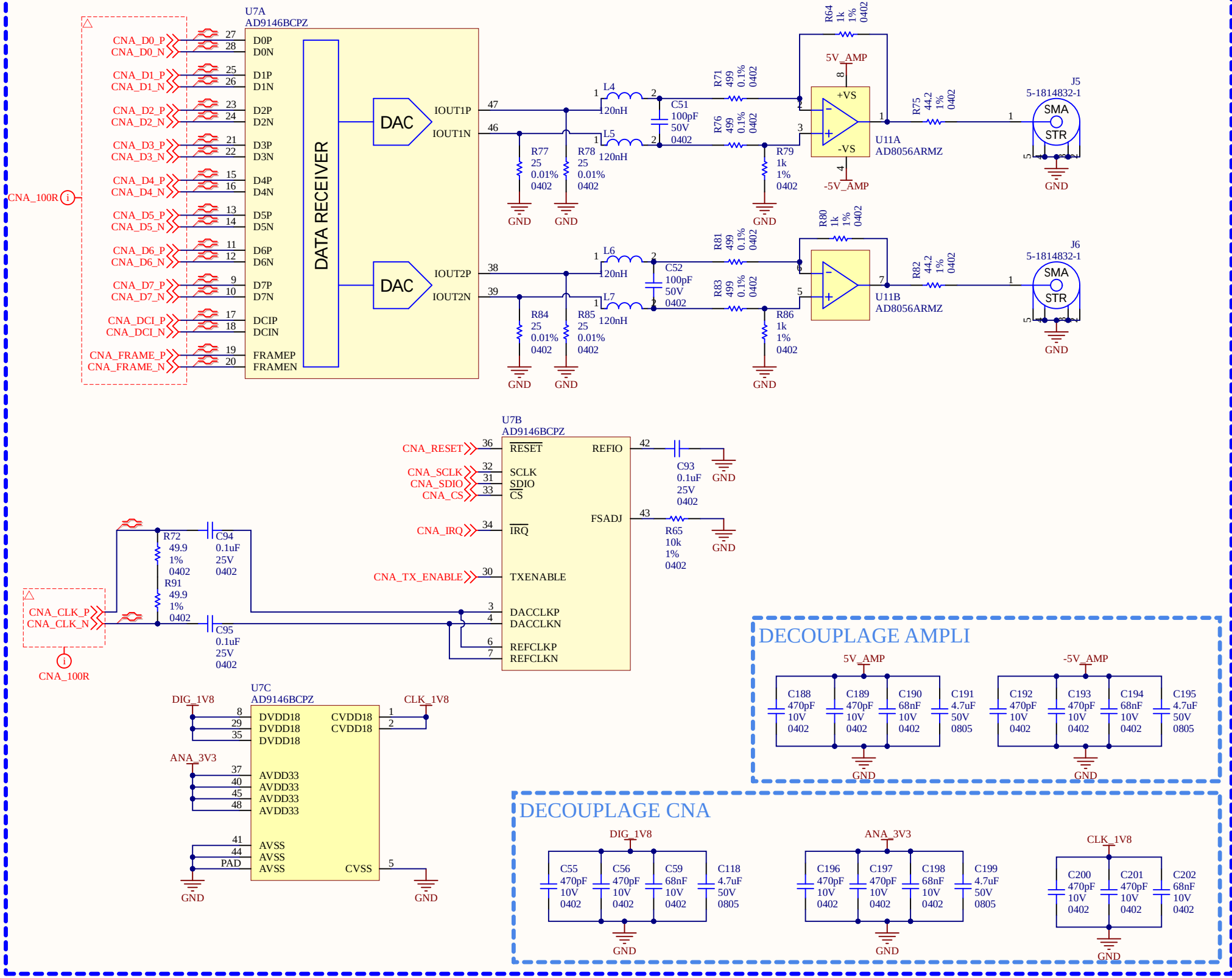
ESD



- Isolation du ETHERNET 1.5kV
- Ajout d'un condensateur de 1.5kV
- Ligne de transmission de 50Ω
- TX et RX voir le référence design pourquoi c'est ligne de transmission sans différentiel

Sheet Name			ETHERNET		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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11x17	01	1.00			
Date	13/01/2026	Sheet	5	of	17
Filename			Designers		
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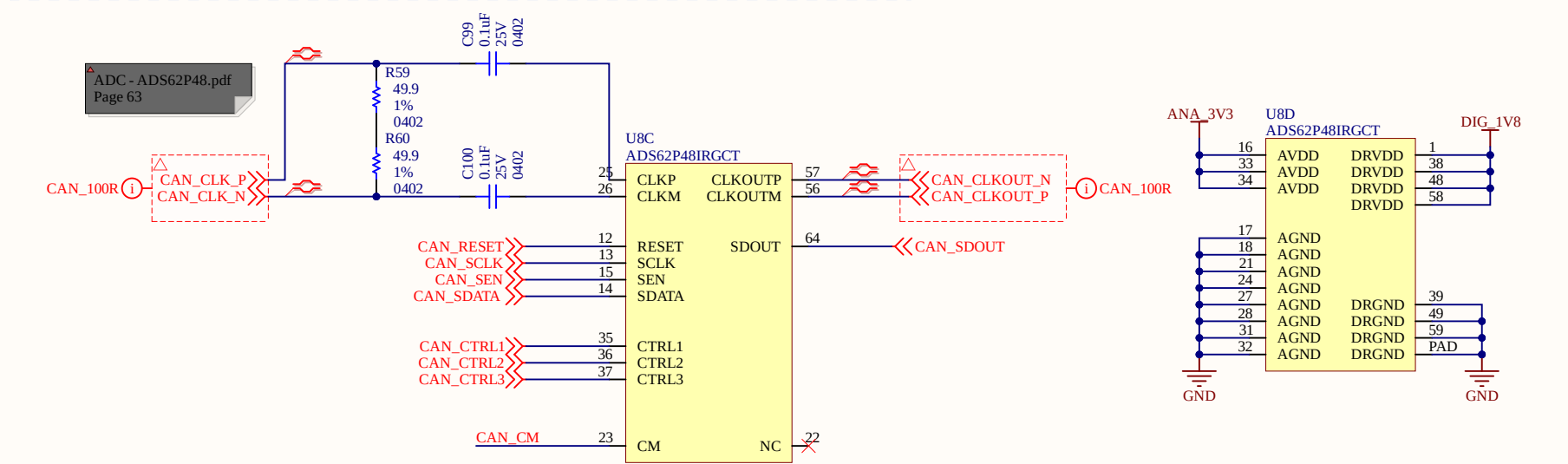
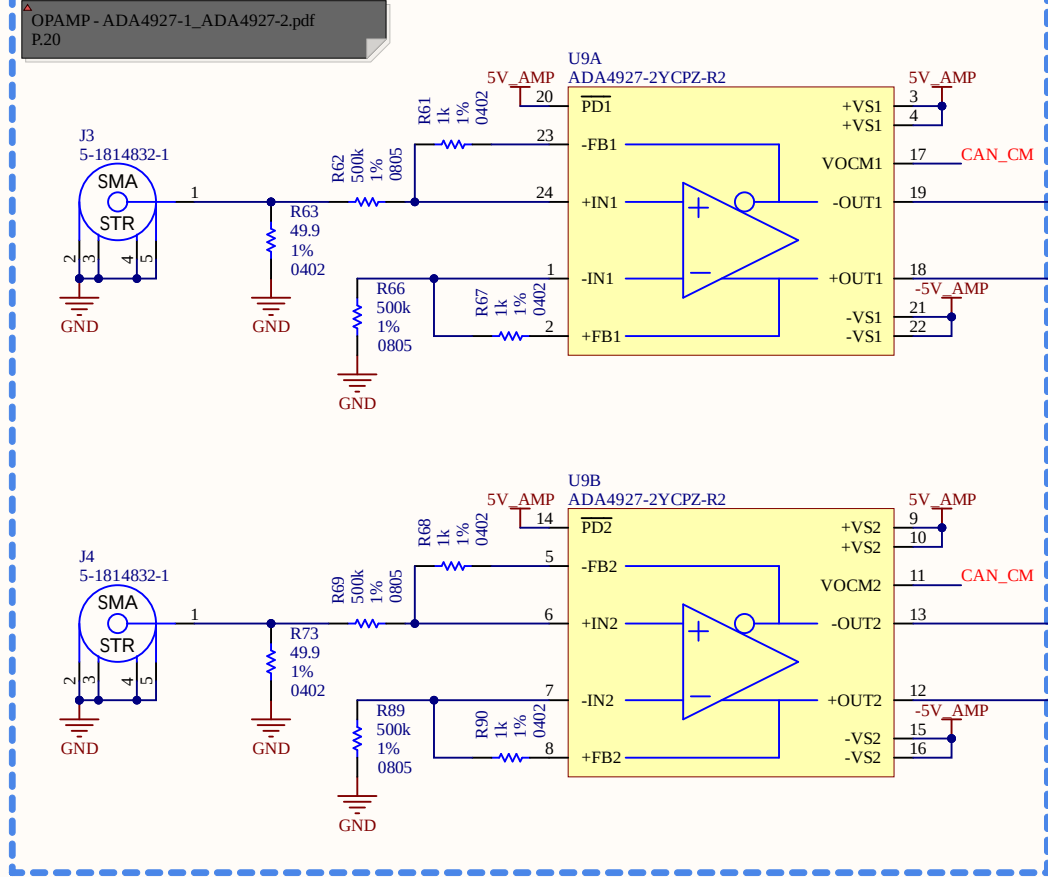
CNA (DAC)



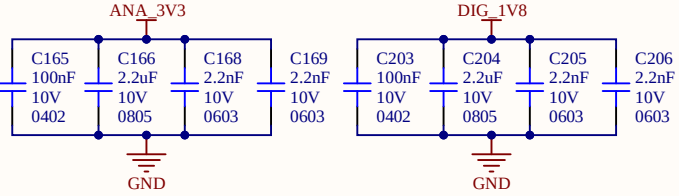
-200mbit/s en DDR LVDS
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Ampli-op aimerait un gain de 2 avec 1V à la sortie, il faut donc 0.5V en entree
-DAC sortie au maximum c'est 20mA, donc avec le 25ohms c'est 0.5V
-On réussi à avoir 1V à la sortie
-5ohms à la sortie à 40MHz donc 45ohms pour adapter
-Filtre passe bas de deuxième ordre (0 à 40MHz), point de coupure 45MHz = L et C
-Receiver Differential impedance 80 to 120ohms

Sheet Name		CNA	
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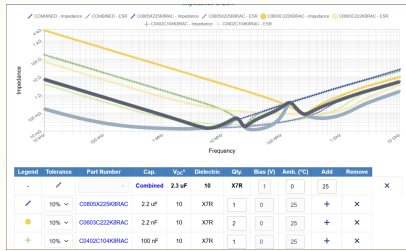
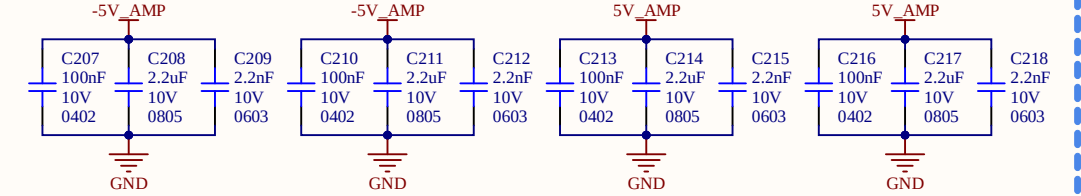
CAN (ADC)
DECOUPLAGE ANALOG



DECOUPLAGE CAN



DECOUPLAGE AMPLI

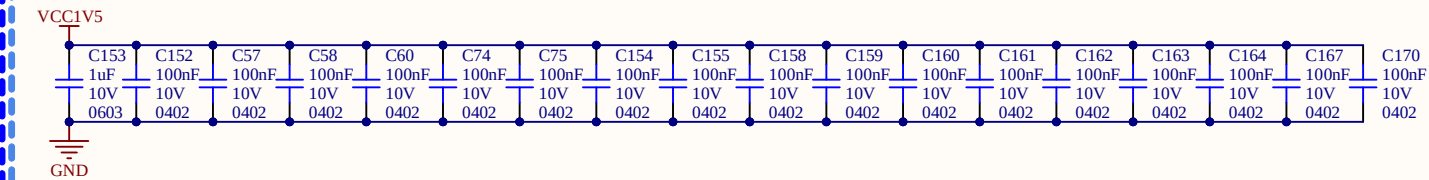


-Choix de OP-AMP pour isolé et pour passer d'analog a digital
-Toute les données vont sur la BANK 1
-Toute le control sur bank 4
- Nous avons pris uniquement AMPLI-OP puisque transfo bloque le DC et on a une composante DC

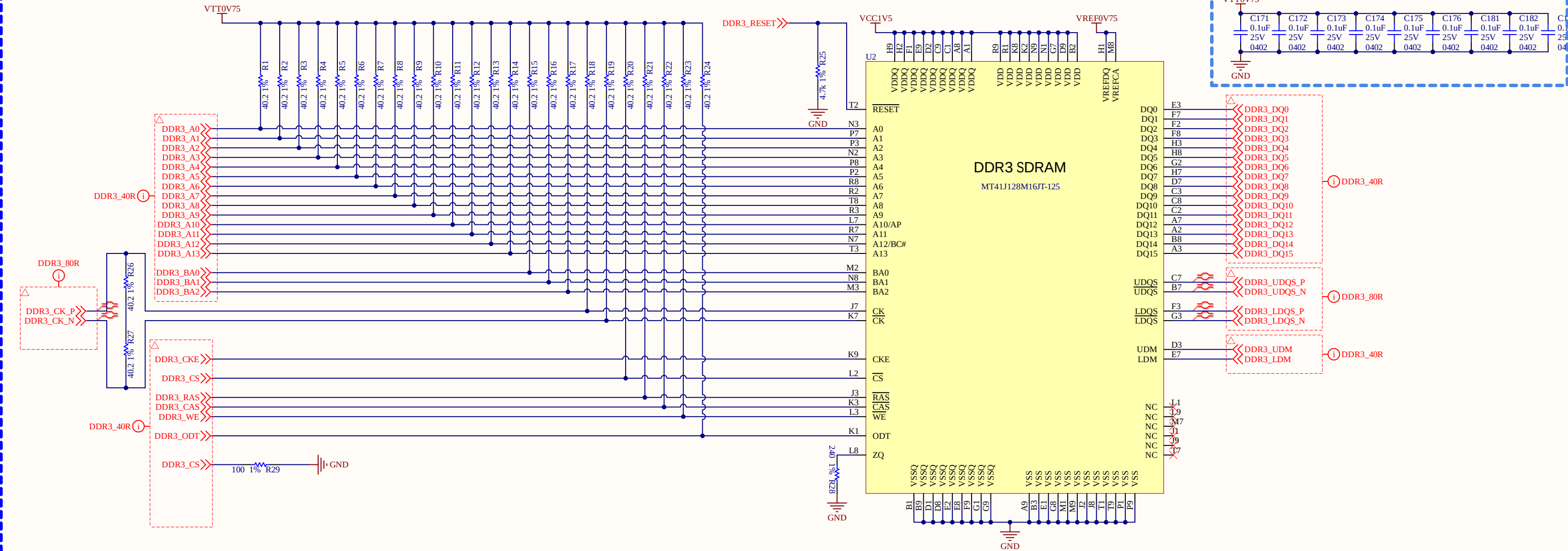
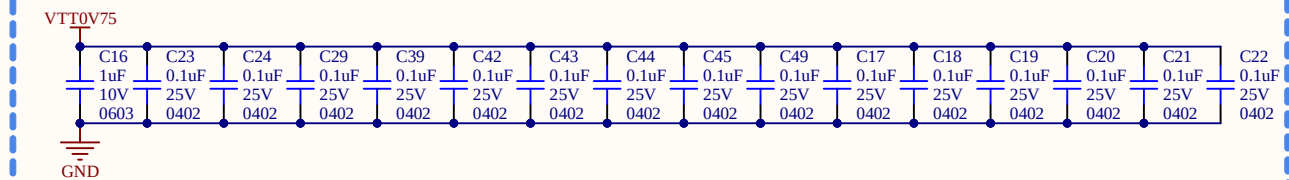
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Global Project			Pleiades		
Size	Group		Revision		
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Date	13/01/2026		Sheet	7	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_CAN.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

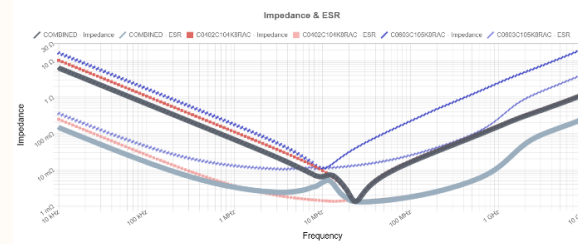
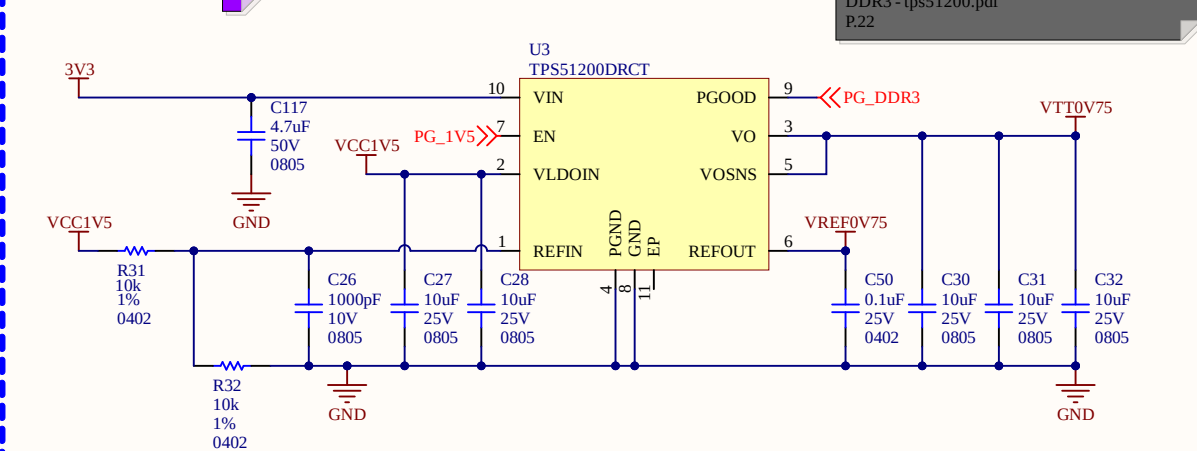
DECOUPLAGE



DECOUPLAGE



ALIM DDR3



- Ligne de transmission de 40R
- Resistance parallele sur la ligne de transmission de 40.2Ohms
- Clock en pair différentiel 80R
- Découplage il y a 18 pins donc on peut en mettre 18 condensateurs
- Dans un cours on s'est fait dire que c'est un condensateur par résistance de VTT

Sheet Name

DDR3

Project Title

S7CAS-PLIEADES-CARTE-MERE

Global Project

Pleiades

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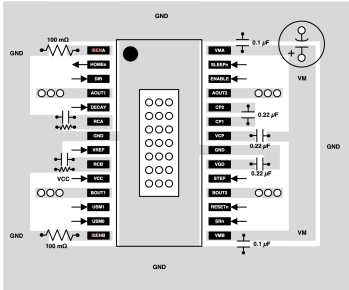
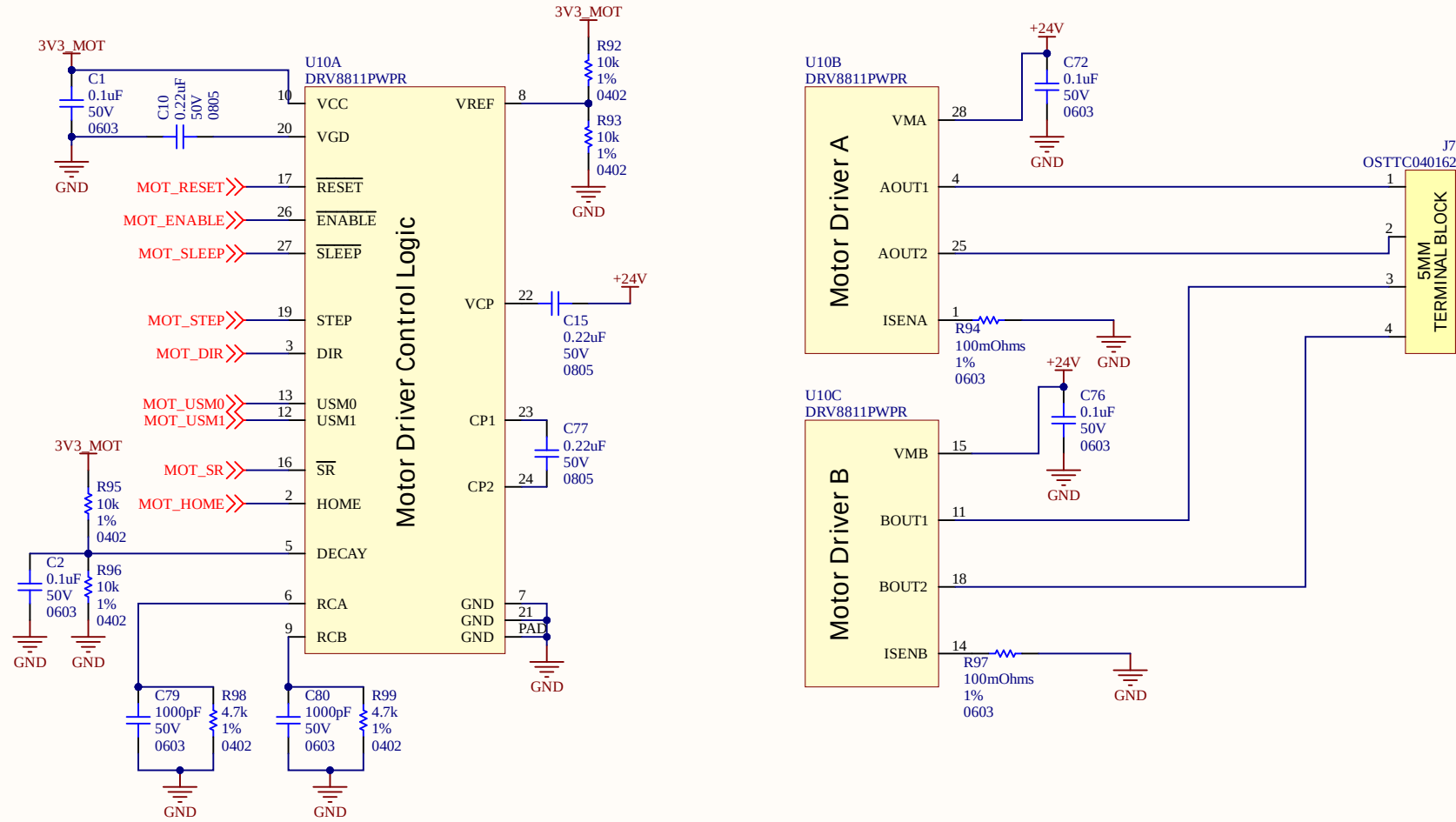
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S7CAS-PLIEADES-CARTE-MERE_P01_DDR3.SchDoc

Designers

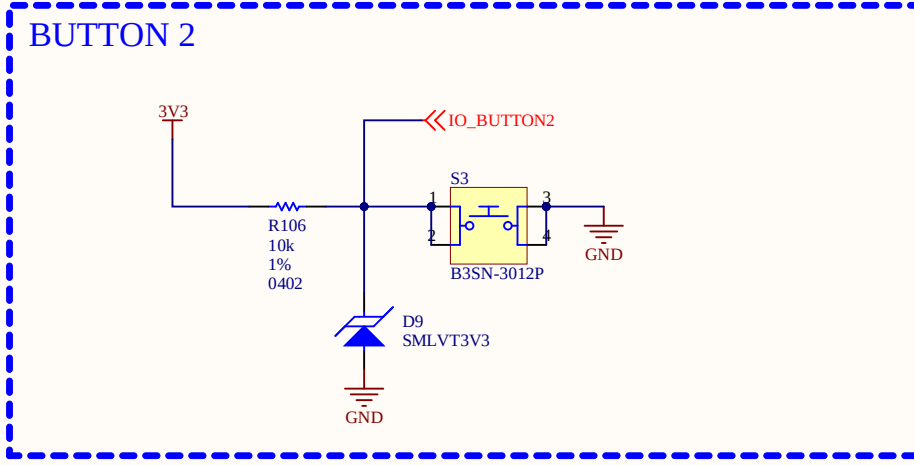
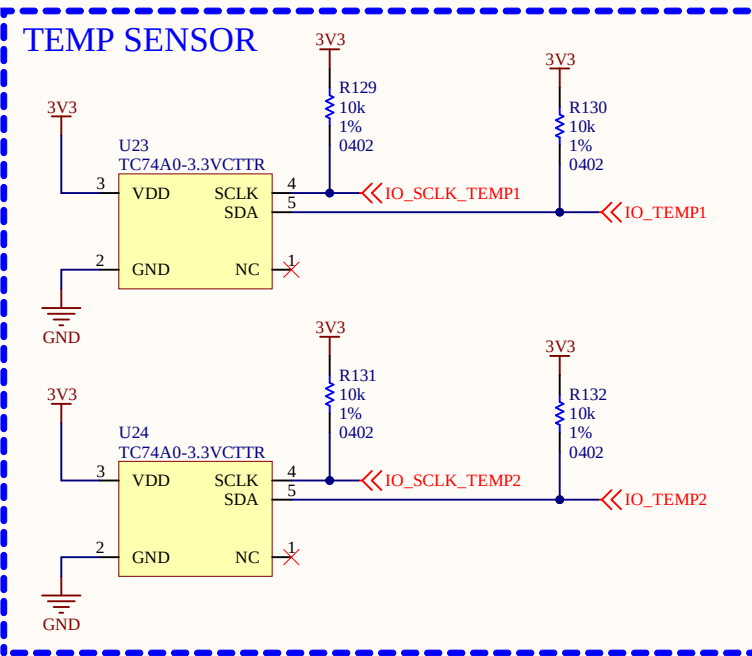
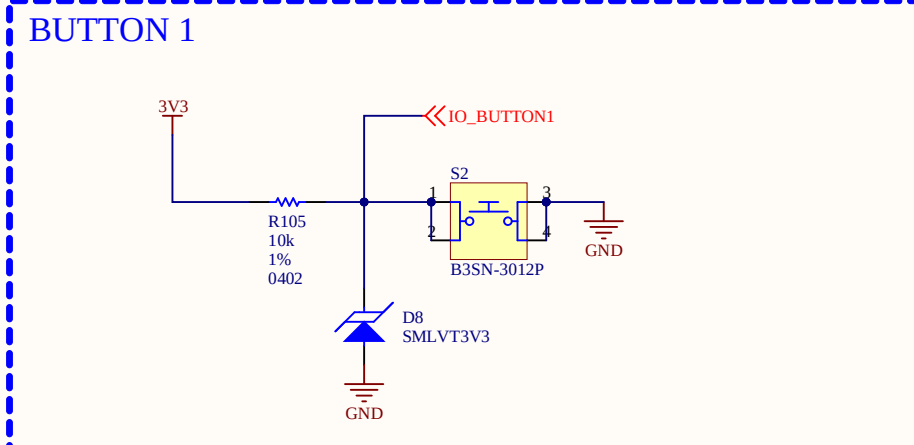
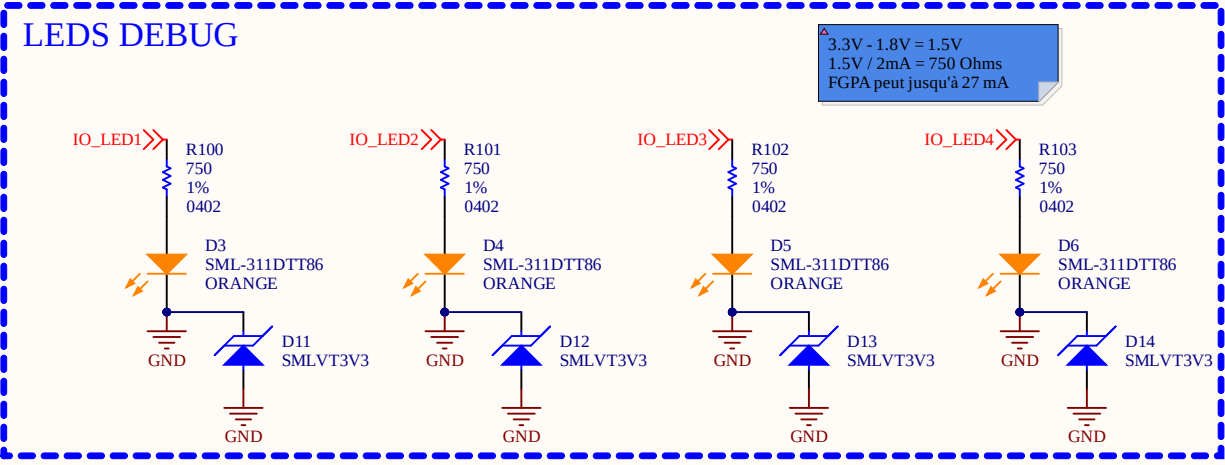
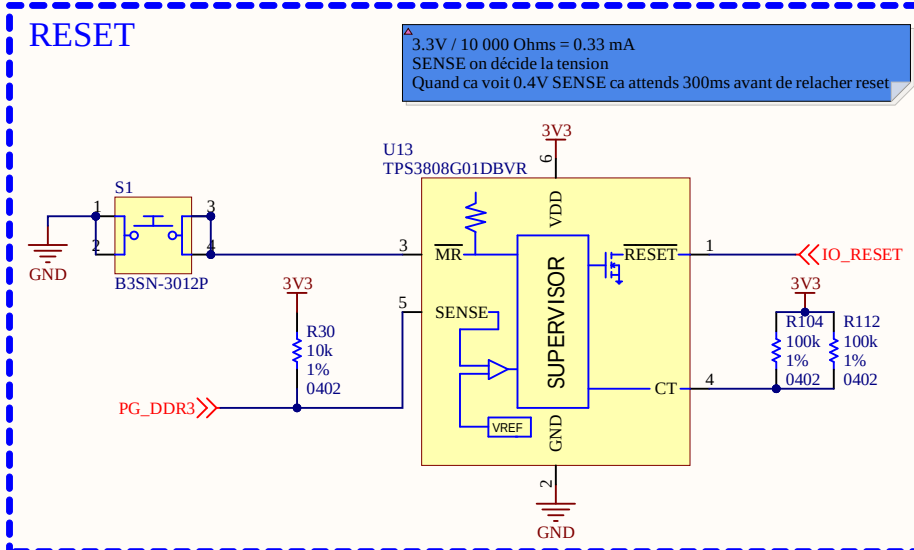
Mathieu Desautels (DESM1210)
Mathys Plante (PLAM1001)
Jonathan Bouchard

MOTOR



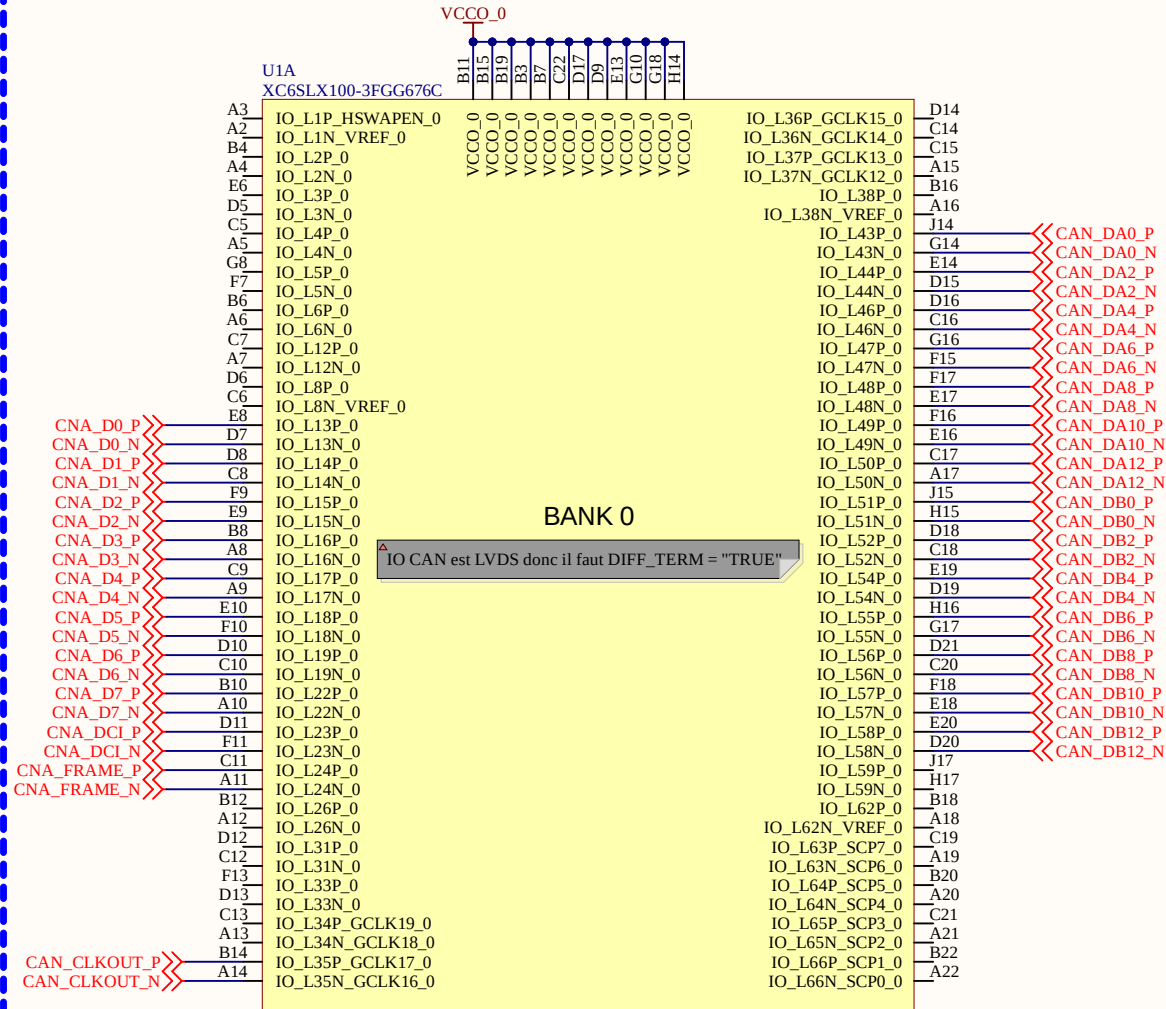
-Tous sur la BANK1
-ISENS = $1.65 / (8 \times 1.9A) = 0,108 \text{ Ohms}$

Sheet Name		MOTORS	
Project Title		S7CAS-PLEIADES-CARTE-MERE	
Global Project		Pleiades	
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11x17	01	1.00	
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Filename		Designers	
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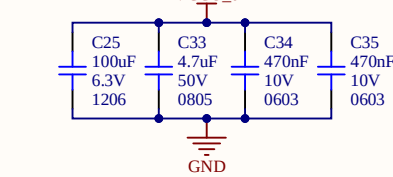


Sheet Name			IO		
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Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	10 of 17	
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_IO.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

FPGA

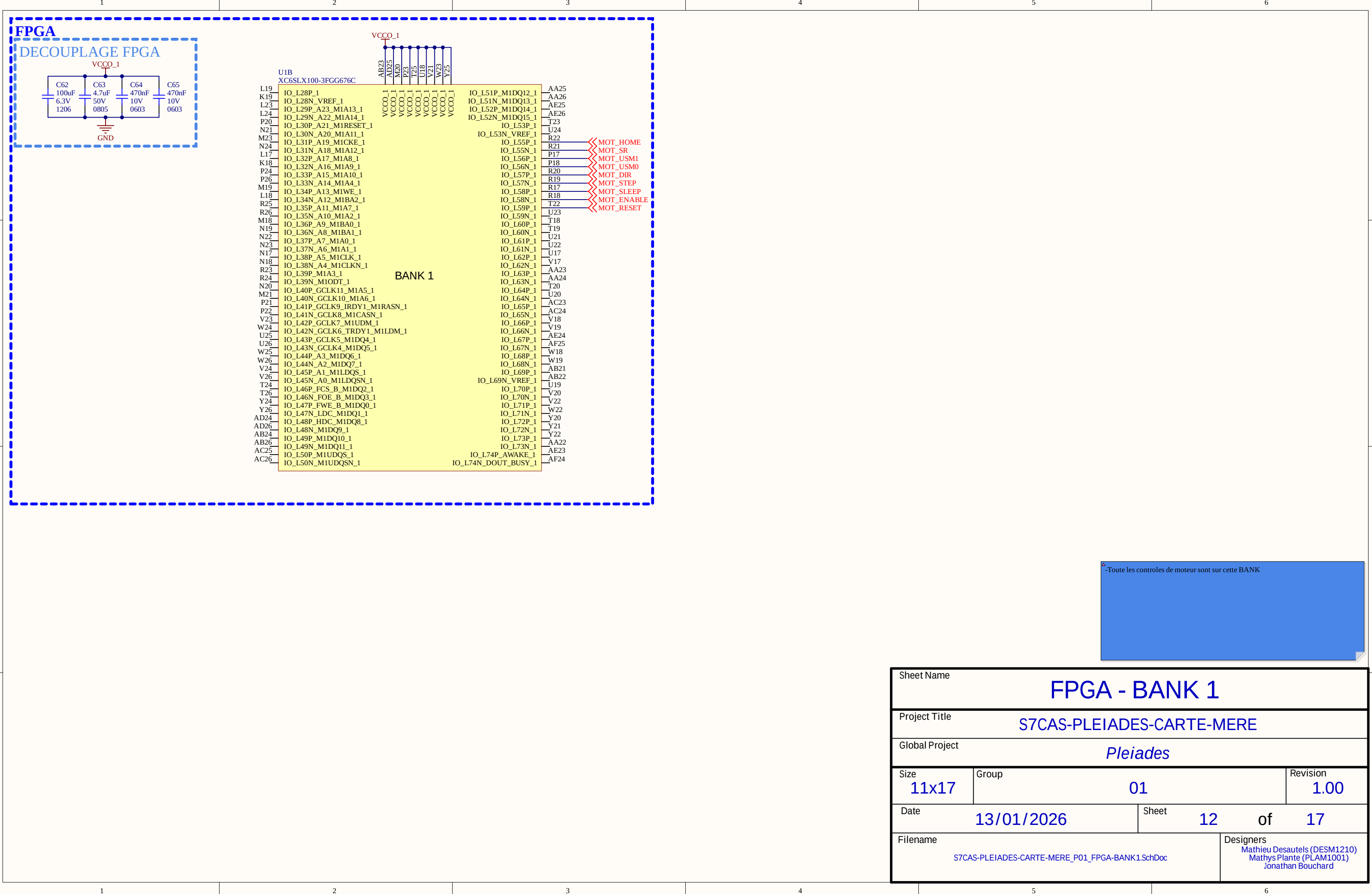


DECOUPLAGE FPGA

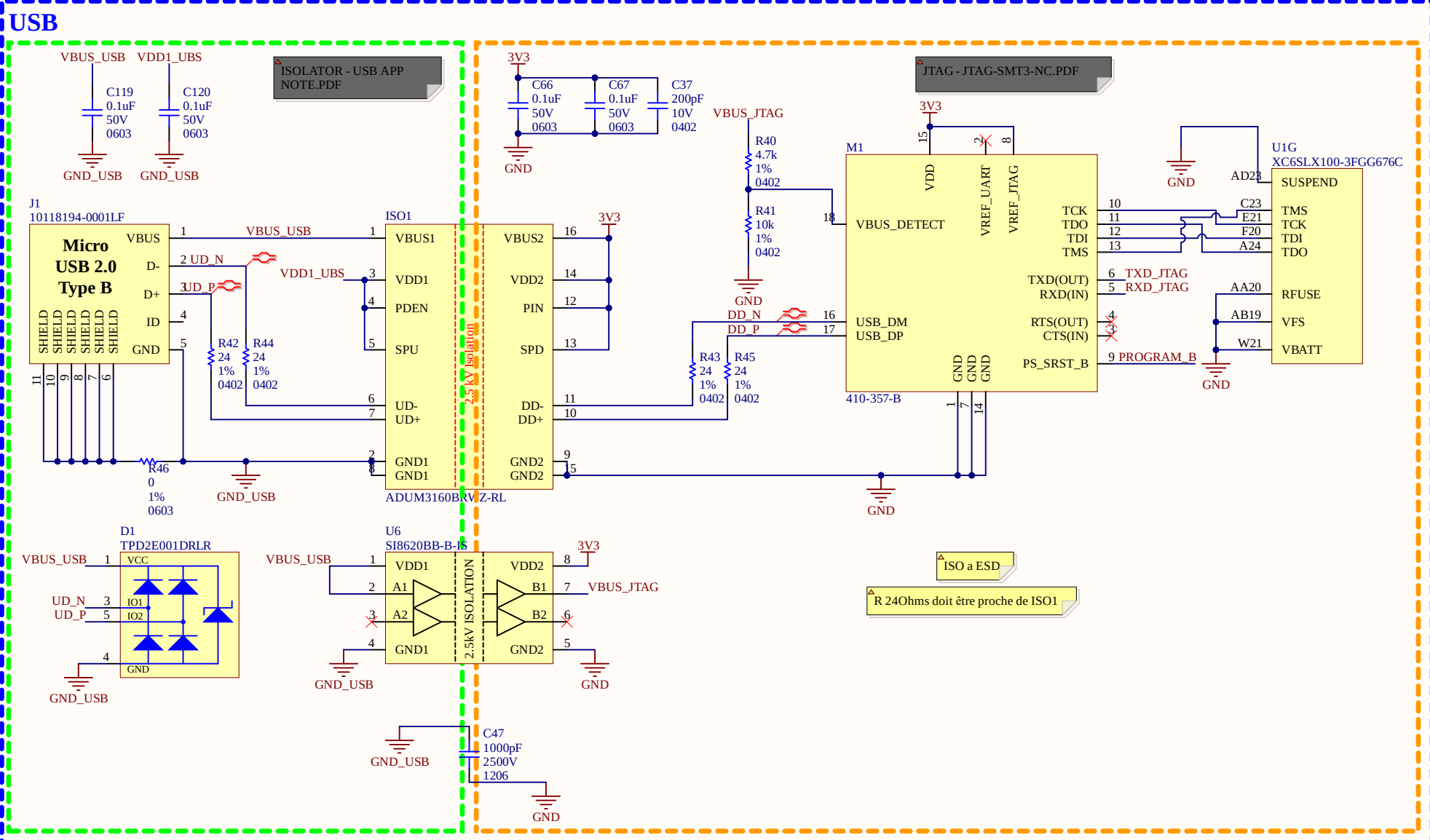
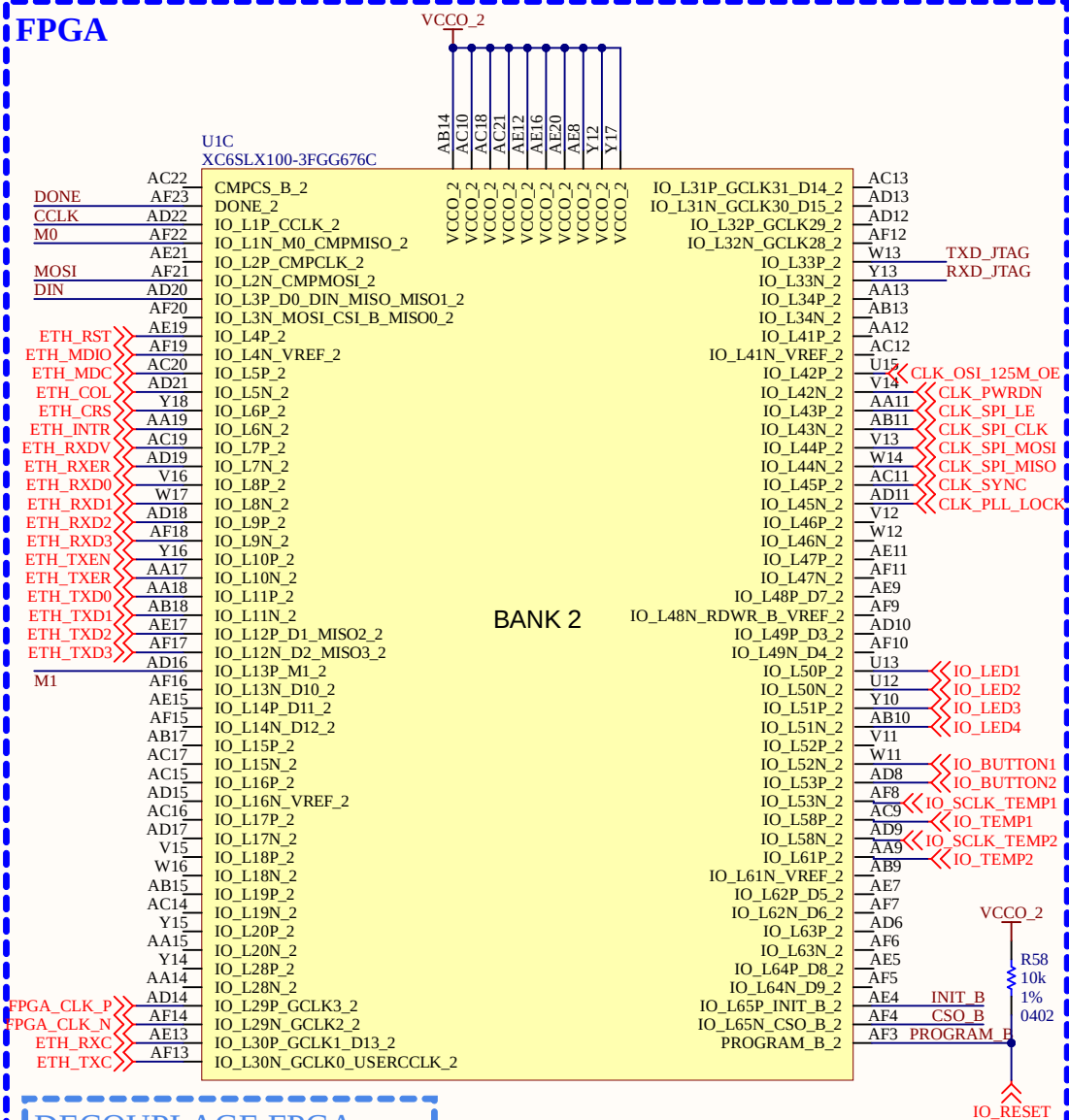


-LVDS seulement sur BANK0 et BANK2
-Toute les CAN et CNA sont mit sur cette BANK

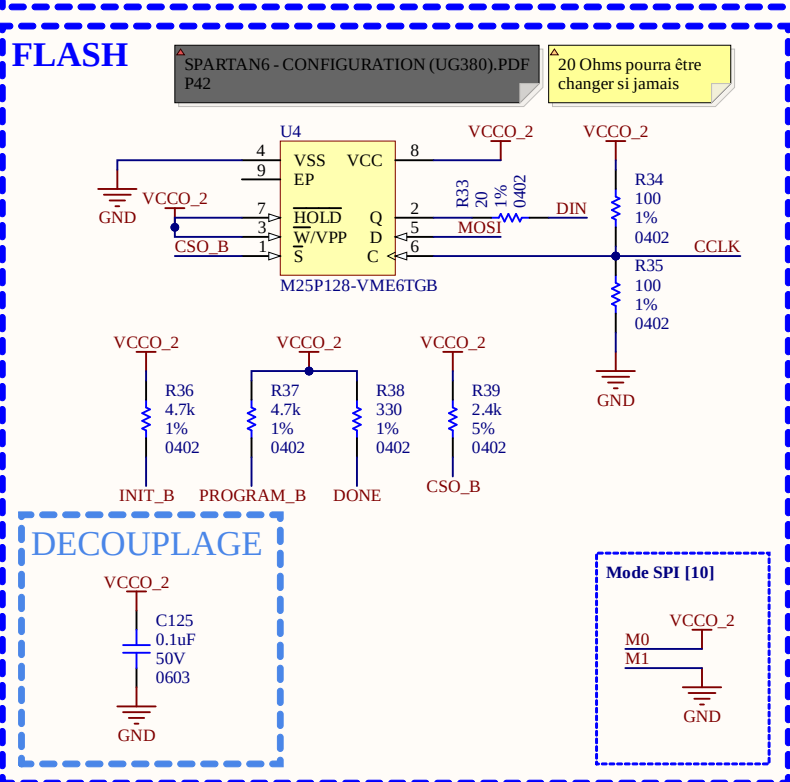
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Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
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S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK0.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		



FPGA



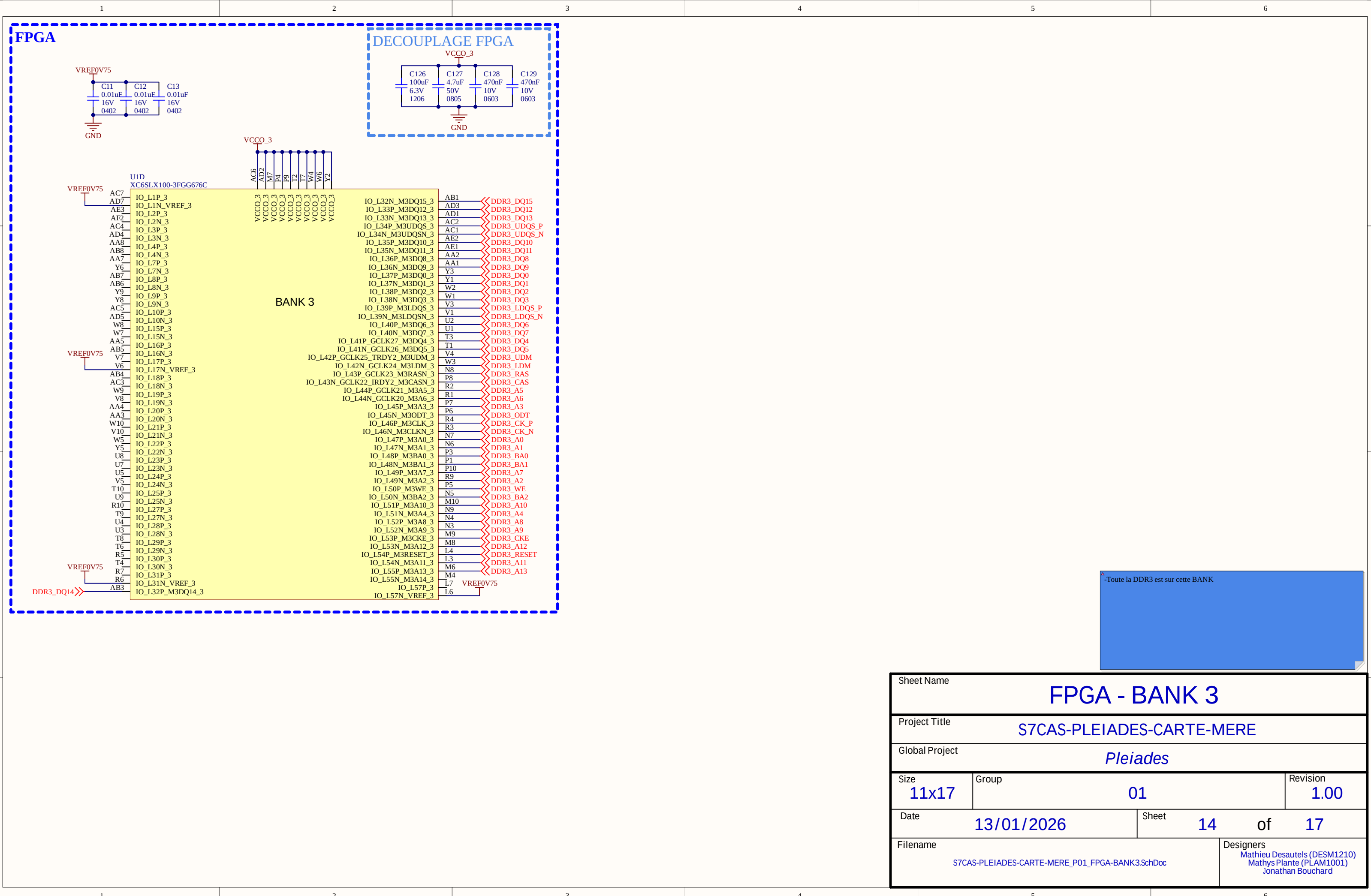
FLASH

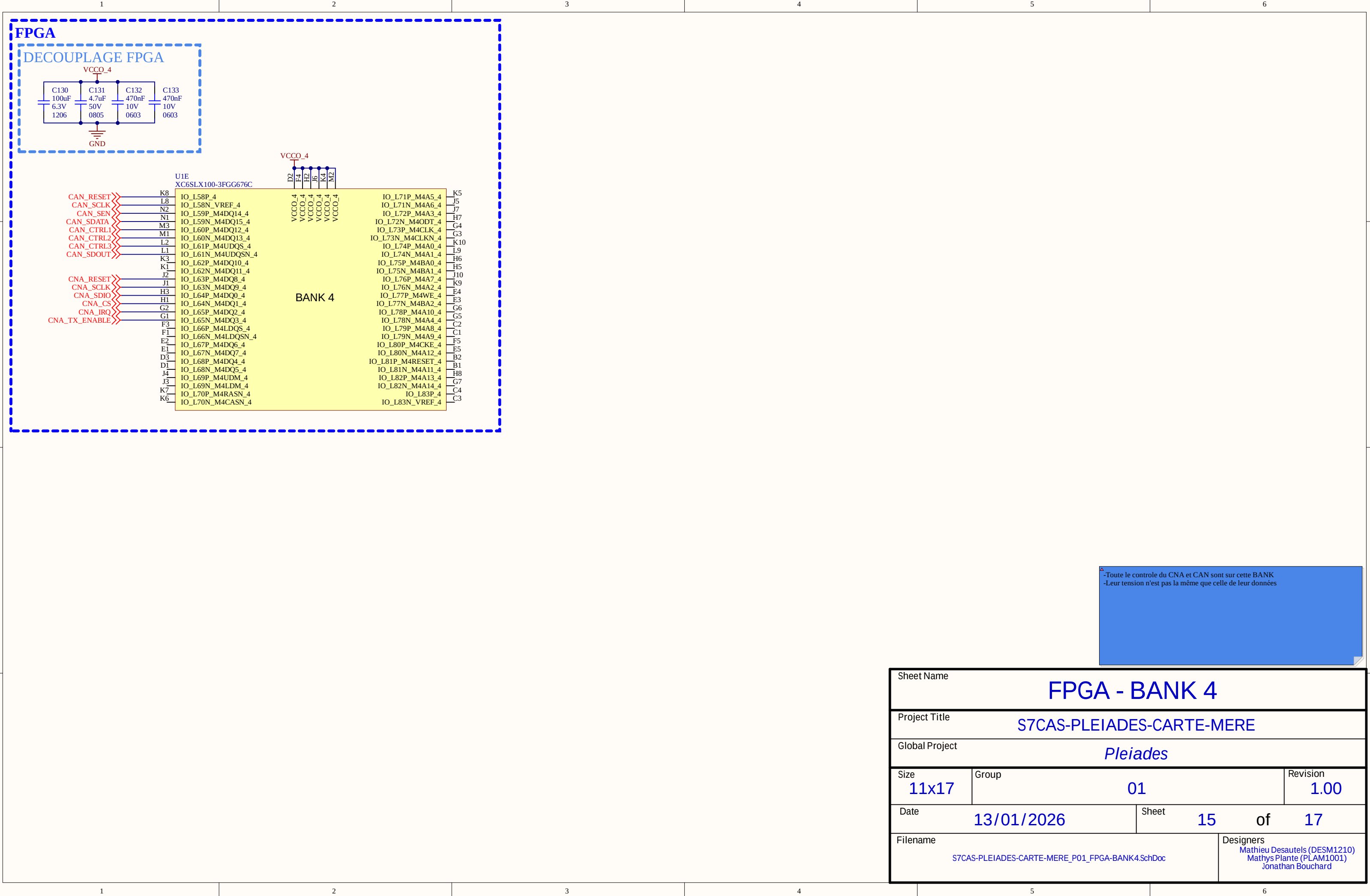


- Ajout de ESD pour la protection
- Protection isolé pour le VBUS
- Condensateur 2.5kV pour isolé les GND (même force que les isolateurs)
- Paires différentiel sur le D+ et D-, avec des résistance de 240ohms (comme l'exemple)
- Isolation du côté USB au lieu d'être du côté board (plus petit comme chip à partager du ISO de 6 fils)
- If AES encryption is unused, VBATT can be tied to either, VCCAUX or ground, or left unconnected.
- VFS and RFUSE are tied to ground because we are not using Efuses

- M0 et M1 choisi pour avoir le mode SPI
- Ajout d'une résistance de 20 ohms sur Q
- Toute la logique de code est sur la BANK ainsi que toute la programmation

Sheet Name			FPGA - BANK 2		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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Size	Group	Revision			
11x17	01	1.00			
Date	Sheet				
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK2.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		





-Toute le controle du CNA et CAN sont sur cette BANK

-Leur tension n'est pas la même que celle de leur données

FPGA

U1F
XC6SLX100-3FGG676C

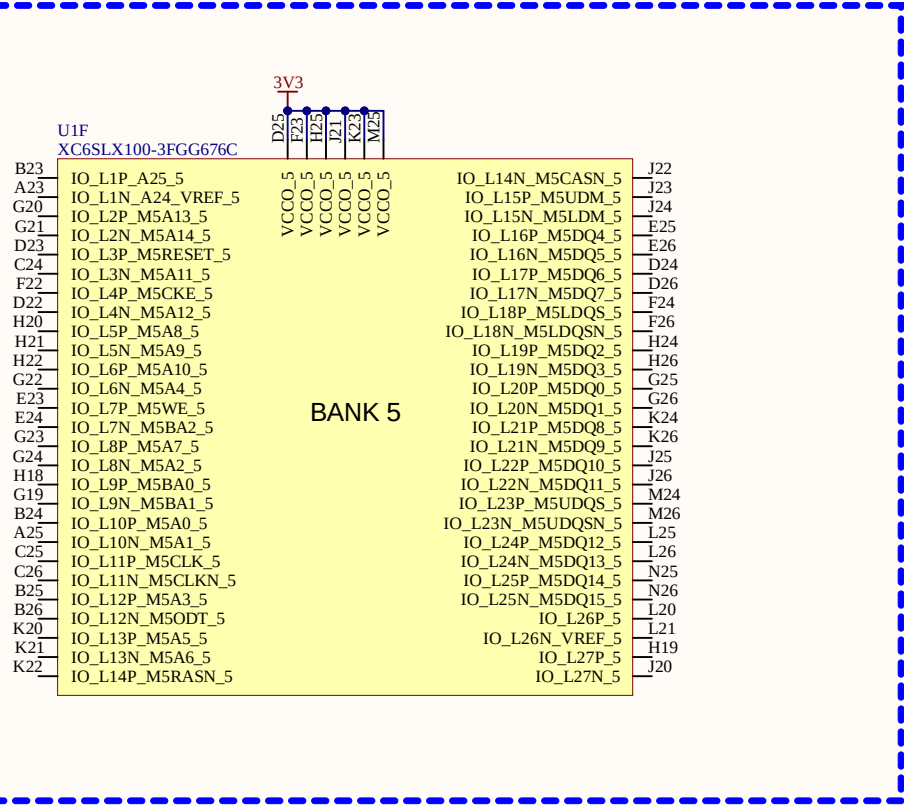
3V3

D25 F23 H23 J21 K23 M25

VCCO_5 VCCO_5 VCCO_5 VCCO_5 VCCO_5

BANK 5

B23	IO_L1P_A25_5	IO_L14N_M5CASN_5	J22
A23	IO_L1N_A24_VREF_5	IO_L15P_M5UDM_5	J23
G20	IO_L2P_M5A13_5	IO_L15N_M5LDM_5	J24
G21	IO_L2N_M5A14_5	IO_L16P_M5DQ4_5	E25
D23	IO_L3P_M5RESET_5	IO_L16N_M5DQ5_5	E26
C24	IO_L3N_M5A11_5	IO_L17P_M5DQ6_5	D24
F22	IO_L4P_M5CKE_5	IO_L17N_M5DQ7_5	D26
D22	IO_L4N_M5A12_5	IO_L18P_M5LDQS_5	F24
H20	IO_L5P_M5A8_5	IO_L18N_M5LDQSN_5	F26
H21	IO_L5N_M5A9_5	IO_L19P_M5DQ2_5	H24
H22	IO_L6P_M5A10_5	IO_L19N_M5DQ3_5	H26
G22	IO_L6N_M5A4_5	IO_L20P_M5DQ0_5	G25
E23	IO_L7P_M5WE_5	IO_L20N_M5DQ1_5	G26
E24	IO_L7N_M5BA2_5	IO_L21P_M5DQ8_5	K24
G23	IO_L8P_M5A7_5	IO_L21N_M5DQ9_5	K26
G24	IO_L8N_M5A2_5	IO_L22P_M5DQ10_5	J25
H18	IO_L9P_M5BA0_5	IO_L22N_M5DQ11_5	J26
G19	IO_L9N_M5BA1_5	IO_L23P_M5UDQS_5	M24
B24	IO_L10P_M5A0_5	IO_L23N_M5UDQSN_5	M26
A25	IO_L10N_M5A1_5	IO_L24P_M5DQ12_5	L25
C25	IO_L11P_M5CLK_5	IO_L24N_M5DQ13_5	L26
C26	IO_L11N_M5CLKN_5	IO_L25P_M5DQ14_5	N25
B25	IO_L12P_M5A3_5	IO_L25N_M5DQ15_5	N26
B26	IO_L12N_M5ODT_5	IO_L26P_5	L20
K20	IO_L13P_M5A5_5	IO_L26N_VREF_5	L21
K21	IO_L13N_M5A6_5	IO_L27P_5	H19
K22	IO_L14P_M5RASN_5	IO_L27N_5	J20



Sheet Name			FPGA - BANK 5		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
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Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-BANK5.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		

A

B

C

D

A

B

C

D

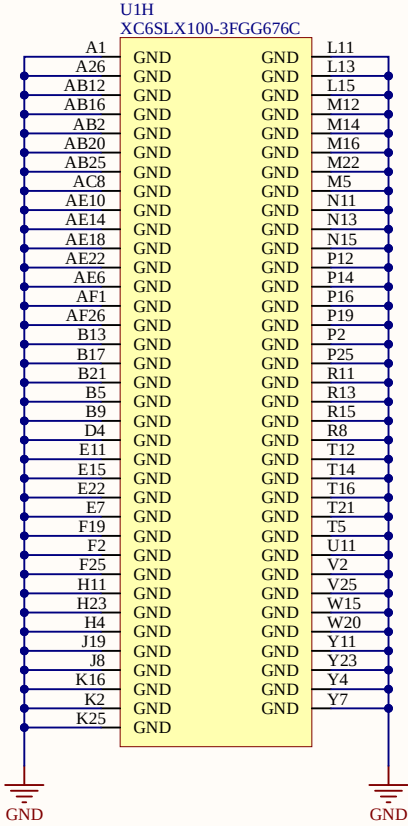
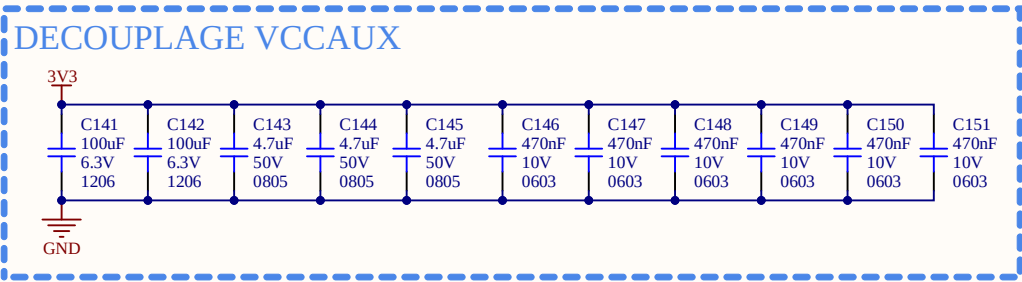
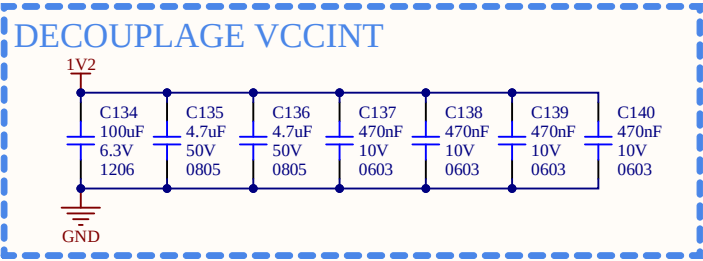
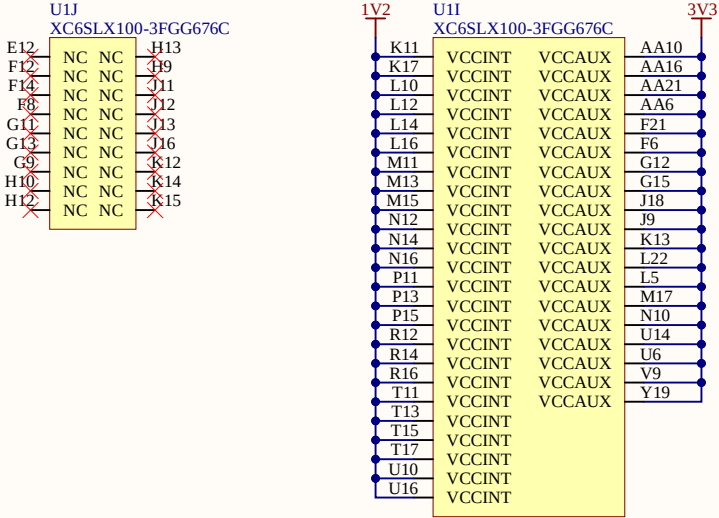


Table 2-1: Required PCB Capacitor Quantities per Device ⁽¹⁾⁽³⁾ (Continued)																	
Package	Device (XCES)	V _{CCINT} In µF	V _{CCAUX} In µF	V _{CC0} Bank 0 In µF	V _{CC0} Bank 1 In µF	V _{CC0} Bank 2 In µF	V _{CC0} Bank 3 In µF	V _{CC0} Bank 4 In µF	V _{CC0} Bank 5 In µF	Total (2)							
		100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	100	4.7	0.47	
FG(G)484-LX75	1 2 3 1 2 4 1 1 2 1 1 3 1 1 4 1 1 3																33
FG(G)484-LX75T	1 2 3 1 2 4 1 1 2 1 1 3 1 1 3 1 1 3																32
FG(G)484-LX100	1 2 4 1 2 4 1 1 2 1 1 3 1 1 3 1 1 3																33
FG(G)484-LX100T	1 2 4 1 2 4 1 1 2 1 1 3 1 1 3 1 1 3																33
FG(G)484-LX150	2 3 6 1 2 4 1 1 2 1 1 3 1 1 3 1 1 3																37
FG(G)484-LX150T	2 3 6 1 2 4 1 1 2 1 1 3 1 1 3 1 1 3																37
FG(G)676-LX45	1 1 2 1 2 5 1 1 3 1 1 3 1 1 3 1 1 4																33
FG(G)676-LX75	1 2 3 2 3 6 1 1 3 1 1 3 1 1 3 1 1 3 1 1 2 1 1 2																45
FG(G)676-LX75T	1 2 3 1 2 5 1 1 3 1 1 2 1 1 3 1 1 2 1 1 2 1 1 2																40
FG(G)676-LX100	1 2 4 2 3 6 1 1 3 1 1 3 1 1 3 1 1 3 1 1 2 1 1 2																46
FG(G)676-LX100T	1 2 4 1 2 5 1 1 3 1 1 2 1 1 3 1 1 3 1 1 2 1 1 2																41
FG(G)676-LX150	2 3 6 2 3 6 1 1 3 1 1 3 1 1 3 1 1 3 1 1 2 1 1 2																50
FG(G)676-LX150T	2 3 6 1 2 5 1 1 3 1 1 2 1 1 3 1 1 2 1 1 2 1 1 2																45
FG(G)900-LX100T	1 2 4 2 3 6 1 1 3 1 1 3 1 1 3 1 1 4 1 1 2 1 1 2																47
FG(G)900-LX150	2 3 6 2 4 7 1 1 5 1 1 3 1 1 5 1 1 4 1 1 2 1 1 2																57
FG(G)900-LX150T	2 3 6 2 3 7 1 1 4 1 1 3 1 1 4 1 1 4 1 1 2 1 1 2																54

Sheet Name			FPGA - ALIM		
Project Title			S7CAS-PLEIADES-CARTE-MERE		
Global Project			Pleiades		
Size	Group		Revision		
11x17	01		1.00		
Date	13/01/2026		Sheet	17	of 17
Filename			Designers		
S7CAS-PLEIADES-CARTE-MERE_P01_FPGA-ALIM.SchDoc			Mathieu Desautels (DESM1210) Mathys Plante (PLAM1001) Jonathan Bouchard		